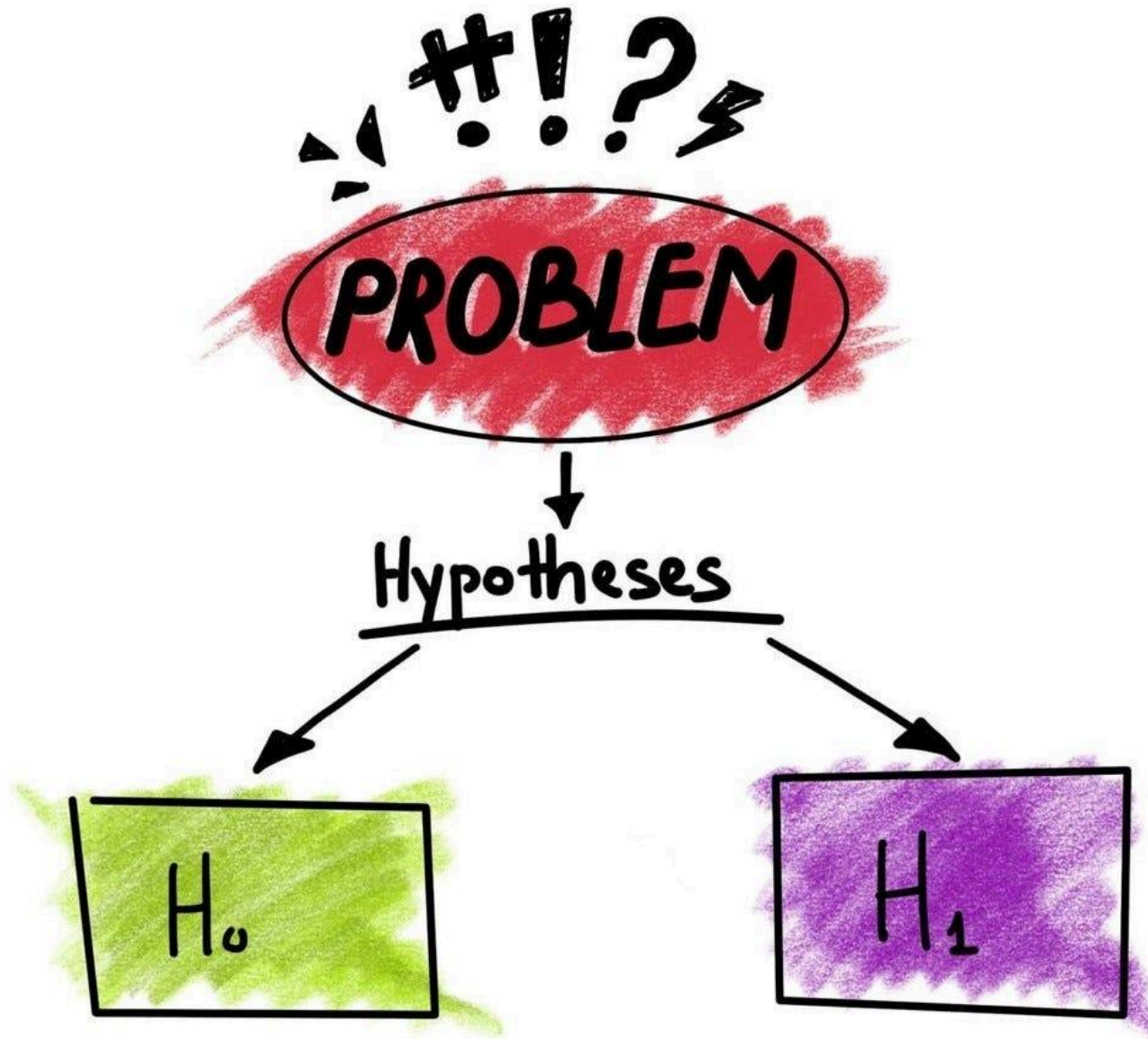




Imagine you work as a Data Scientist for Amazon.

The product team says,

"We changed the Buy Now button from blue to green."

Now they ask,

"Did the conversion rate actually increase?"

Can we simply compare two numbers?

Old Version = 5%

New Version = 5.3%

Is the improvement real?

Or did it happen by chance?

This is where **Hypothesis Testing** comes into the picture.

It helps us determine whether an observed difference is statistically significant or simply due to random variation.

Why Hypothesis Testing?

Real-world decisions involve uncertainty.

Examples:

- Did a new marketing campaign increase sales?
- Does a new recommendation algorithm improve user engagement?
- Is a new machine learning model significantly better than the existing one?
- Has customer satisfaction improved after introducing a chatbot?
- Does a medicine reduce blood pressure?

Rather than relying on intuition, hypothesis testing uses data to make evidence-based decisions.

What is a Statistical Hypothesis?

A statistical hypothesis is an assumption about a population parameter that can be tested using sample data.

Examples:

- Average salary = ₹50,000
- Mean delivery time = 2 days
- Conversion rate = 10%
- Average customer rating = 4.5

Since collecting data from an entire population is often impractical, we draw a sample and test whether the assumption is supported.

What is the Null Hypothesis (H_0)?

The Null Hypothesis represents the default assumption.

It assumes:

- No change
- No difference
- No relationship
- No improvement
- No effect

It is considered true unless there is sufficient evidence to reject it.

Example 1

Current conversion rate = 10%

After redesign:

H_0 :

The conversion rate remains 10%.

There is no improvement.

Example 2

A company launches a new recommendation system.

H_0 :

The new recommendation engine performs the same as the old one.

Example 3

Machine Learning

Current model accuracy = 90%

New model accuracy = 91%

H_0 :

The increase is due to random chance.

The new model is **not significantly better**.

What is the Alternate Hypothesis (H_1 or H_a)?

The Alternate Hypothesis represents what we want to prove.

It states that there is:

- A difference
- An improvement
- A relationship
- An effect

Example

Old Website Conversion

10%

New Website

12%

H_1 :

The new website increases the conversion rate.

Machine Learning Example

Old Accuracy

90%

New Accuracy

94%

H_1 :

The new model performs significantly better.

Understanding H_0 and H_1 Together

Scenario	H_0	H_1
Website Conversion	No increase	Conversion increased
Sales	Sales unchanged	Sales increased
Customer Satisfaction	No improvement	Satisfaction improved
Delivery Time	Same delivery time	Delivery time reduced
ML Accuracy	No significant improvement	Significant improvement

Courtroom Analogy

A useful way to understand hypothesis testing is through a courtroom example.

Person on Trial

Null Hypothesis (H_0): The person is innocent.

Alternate Hypothesis (H_1): The person is guilty.

The court assumes innocence unless there is strong evidence to prove guilt.

Similarly, in hypothesis testing, we assume the null hypothesis is true until the data provides enough evidence to reject it.

One-Tailed vs Two-Tailed Hypothesis

Right-Tailed Test

We only want to know whether the value increased.

Example:

Did sales increase?

H_0 :

Sales $\leq$ Previous sales

H_1 :

Sales $>$ Previous sales

Left-Tailed Test

We only want to know whether the value decreased.

Example:

Did delivery time reduce?

H_0 :

Delivery time $\geq$ Old time

H_1 :

Delivery time $<$ Old time

Two-Tailed Test

We only want to know whether the value changed, regardless of direction.

Example:

Did the average salary change?

H_0 :

Mean salary = ₹50,000

H_1 :

Mean salary $\neq$ ₹50,000

Business Examples

Netflix

Question:

Did the new recommendation algorithm increase watch time?

H_0 :

No increase.

H_1 :

Watch time increased.

Amazon

Question:

Did changing the "Buy Now" button improve sales?

H₀:

No improvement.

H₁:

Sales improved.

Uber

Question:

Did reducing prices increase bookings?

H₀:

Bookings remained the same.

H₁:

Bookings increased.

Hospital

Question:

Did the medicine lower blood pressure?

H_0 :

No reduction.

H_1 :

Blood pressure decreased.

Type I and Type II Errors

When making decisions based on sample data, mistakes can occur.

Type I Error (False Positive)

Rejecting the null hypothesis when it is actually true.

Example:

You conclude the new website improves conversions, but in reality it does not.

Business impact:

The company spends money implementing a feature that provides no real benefit.

Type II Error (False Negative)

Failing to reject the null hypothesis when the alternative hypothesis is actually true.

Example:

The new recommendation algorithm genuinely improves engagement, but the test concludes there is no improvement.

Business impact:

The company misses an opportunity to improve performance.

Reality	Decision	Error
H_0 True	Reject H_0	Type I Error
H_1 True	Fail to Reject H_0	Type II Error

Applications in Data Science & Machine Learning

Feature Selection

Determine whether a feature is significantly related to the target variable.

Example:

Is "Income" significantly associated with loan approval?

Model Comparison

Compare two machine learning models to determine if one performs significantly better.

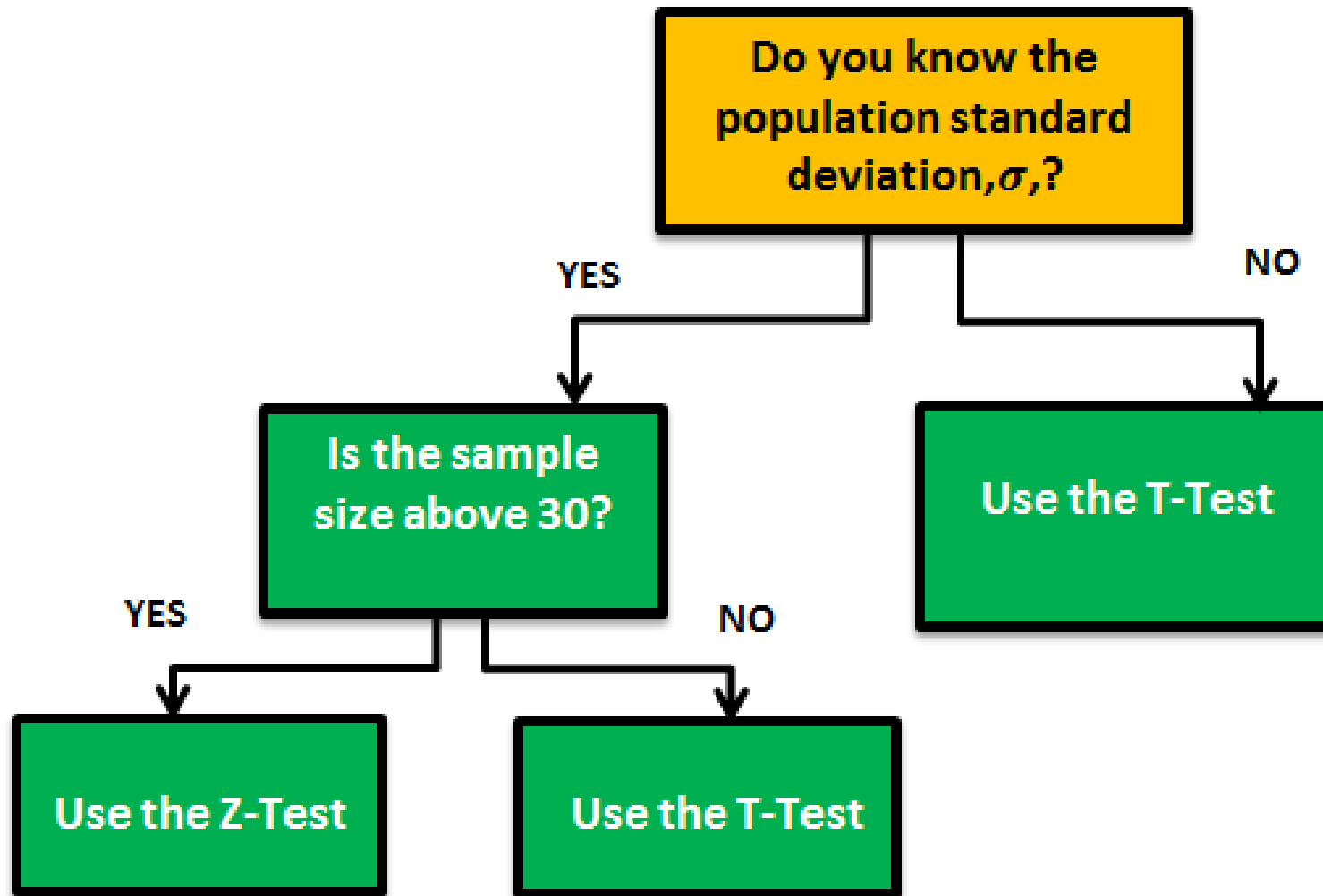
Example:

Random Forest vs XGBoost

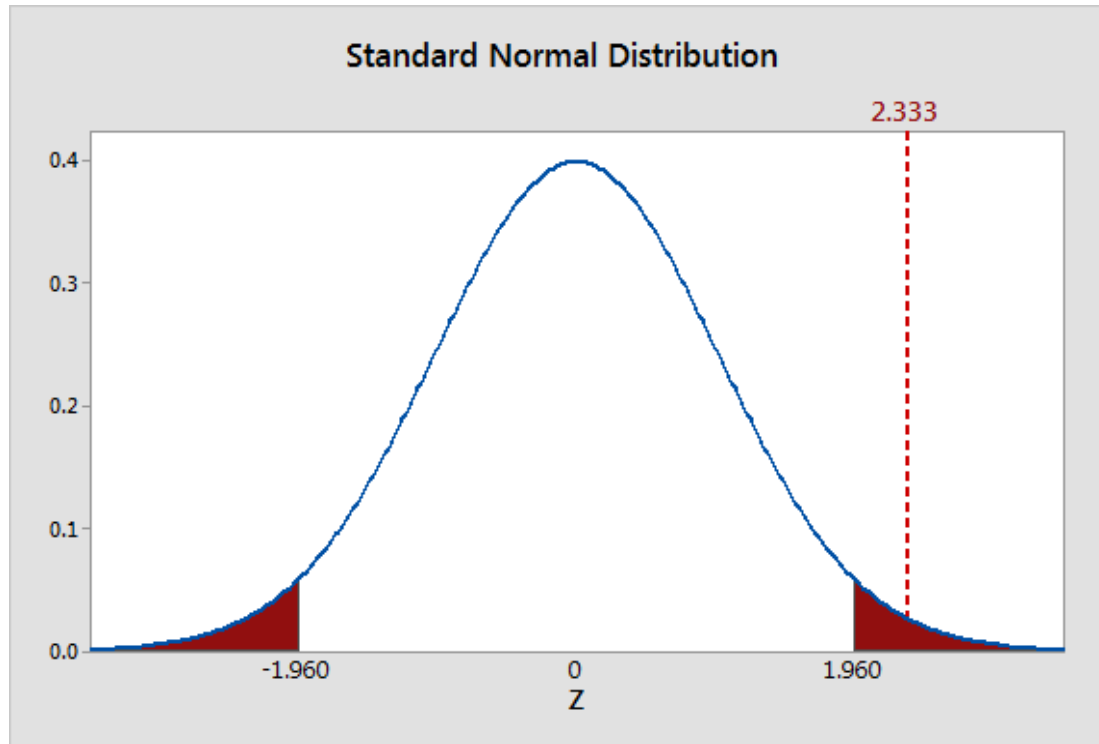
Business Analytics

- Marketing campaign effectiveness
- Customer retention
- Fraud detection
- Pricing strategy
- Product recommendation systems

Z-Test and T-Test



What is a Z-Test?



A **Z-test** is a statistical hypothesis test used to determine whether the difference between a sample statistic and a population parameter (or between two sample statistics) is statistically significant.

It measures **how many standard errors** the observed value is away from the expected value under the null hypothesis.

When Do We Use a Z-Test?

A Z-test is appropriate when:

- Sample size is **large** ($n \geq 30$).
- Population standard deviation (σ) is known, or the sample is large enough for the Central Limit Theorem to apply.
- Samples are randomly selected.
- Observations are independent.
- The sampling distribution of the mean is approximately normal.

Types of Z-Tests

A. One-Sample Z-Test

Used to compare a sample mean with a known population mean.

Example:

- Population average salary = ₹50,000
- Sample average salary = ₹51,200

Question:

- Is the salary significantly different?

B. Two-Sample Z-Test

Compares the means of two independent groups.

Example:

- Old Website Conversion :8%
- New Website Conversion : 9%

Question:

- Did the new website improve conversions?

C. Z-Test for Proportions

Used to compare proportions instead of means.

Example:

- Old advertisement : 5% Click Through Rate
- New advertisement : 7% Click Through Rate

Used heavily in:

- A/B Testing
- Marketing Analytics
- Product Analytics

Z-Score

The Z-score indicates how far a value is from the expected value in terms of standard errors.

A higher absolute Z-score means the observed result is less likely to have occurred by chance.

General interpretation:

- **Z = 0** → Exactly matches the expected value.
- **Positive Z** → Above the expected value.
- **Negative Z** → Below the expected value.
- **Large |Z|** → Stronger evidence against the null hypothesis.

Formula

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}$ = Sample mean
- μ = Population mean
- σ = Population standard deviation
- n = Sample size

Step 1

- Difference
- Sample Mean – Population Mean
- Measures how far the sample mean is from the population mean.

Step 2

- Standard Error
- $SE = \frac{\sigma}{\sqrt{n}}$
- Measures how much the sample mean is expected to vary from sample to sample.

Step 3

- Z-score
- Difference ÷ Standard Error
- This tells us whether the observed difference is unusually large.

Example

A beverage company claims:

Average bottle fill = **500 ml**

Sample information:

- Sample size = **100**
- Sample mean = **503 ml**
- Population standard deviation = **10 ml**
- Significance level = **0.05**

What is the Significance Level ($\alpha = 0.05$)?

The significance level (α) is a threshold that determines how much evidence is required to reject the Null Hypothesis (H_0).

It represents the **maximum probability of making a Type I Error**, which means rejecting the null hypothesis when it is actually true.

Relationship Between α and Confidence Level

The confidence level is simply:

Confidence Level = $1 - \alpha$

α	Confidence Level
0.10	90%
0.05	95%
0.01	99%

So,

$\alpha = 0.05$ means a 95% confidence level.

Step 1

Hypotheses

H_0 :

- Average fill = 500 ml

H_1 :

- Average fill $\neq$ 500 ml

Step 2

Calculate Standard Error

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$$

Step 3

Calculate Z-score

$$Z = \frac{503 - 500}{1} = 3$$

Step 4

Decision

- For a two-tailed test at $\alpha = 0.05$:
- Critical Z-values = ± 1.96

Since:

- $3 > 1.96$
- Reject H_0 .

Z-Test Practice Question

Question:

The average height of all residents in a city is 168 cm, with a population standard deviation of 3.9 cm.

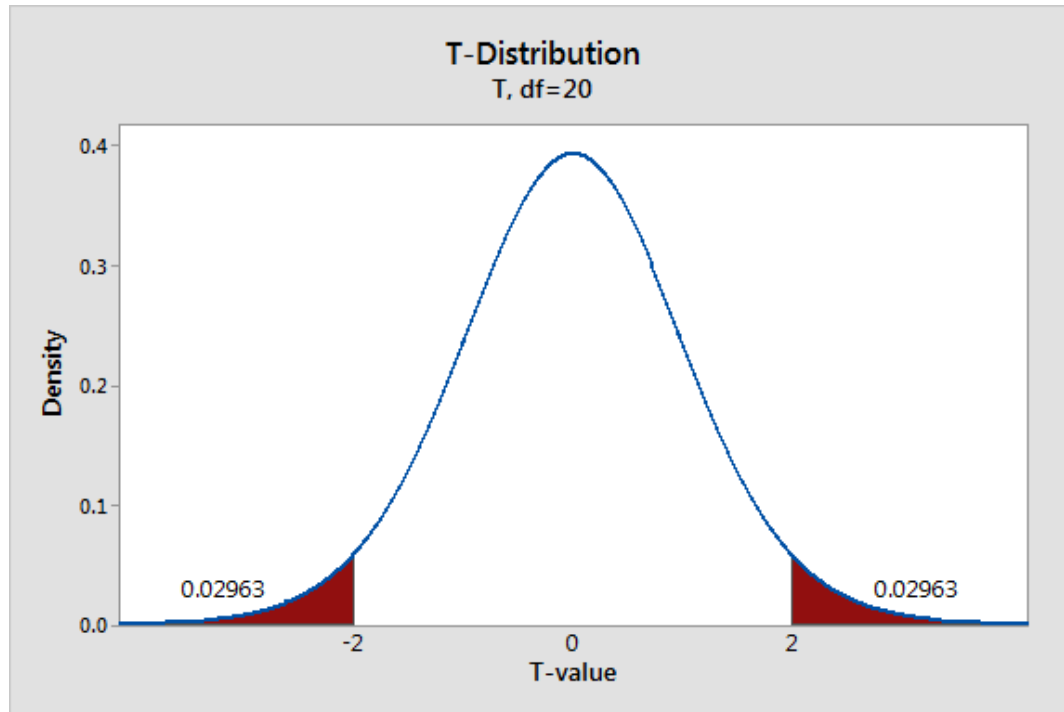
A doctor believes that the average height of the city's residents is different from 168 cm. To investigate this, the doctor randomly measures the heights of 36 individuals and finds that their sample mean height is 169.5 cm.

Using a **95% confidence level ($\alpha = 0.05$)**,

State the **Null Hypothesis (H_0)** and **Alternative Hypothesis (H_1)**.

Write the final conclusion in the context of the problem: **Is there sufficient evidence to conclude that the average height of the city's residents is different from 168 cm?**

What is a T-Test?



A **T-test** is a statistical hypothesis test used to compare means when:

- The sample size is small (typically $n < 30$), and/or
- The population standard deviation (σ) is unknown.

It uses the **Student's t-distribution**, which accounts for the extra uncertainty introduced by estimating the population standard deviation from the sample.

Why Do We Need a T-Test?

In many real-world situations:

- We have only a small sample.
- The population standard deviation is unknown.
- We still want to determine whether the observed difference is statistically significant.

The T-test helps us make this decision.

When Do We Use a T-Test?

Use a T-test when:

- Sample size is **less than 30**.
- Population standard deviation is **unknown**.
- Data is approximately normally distributed.
- Samples are randomly selected.
- Observations are independent.

Types of T-Tests

A. One-Sample T-Test

Compares the mean of one sample with a known or hypothesized population mean.

Example:

A company claims the average battery life is **10 hours**.

You test **20 batteries**.

Is the average battery life actually 10 hours?

B. Independent Two-Sample T-Test

Compares the means of two independent groups.

Example:

- Old teaching method
- New teaching method

Question:

Do students taught using the new method score significantly higher?

C. Paired T-Test

Compares two related measurements taken from the same subjects.

Examples:

- Before and after a fitness program
- Before and after a medical treatment
- Employee productivity before and after training

Formula

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Where:

- $\bar{x}$ = Sample mean
- μ = Population mean
- σ = Population standard deviation
- n = Sample size

Step 1

- Difference
- Sample Mean – Population Mean
- Measures the observed difference.

Step 2

- Standard Error
- $SE = \frac{s}{\sqrt{n}}$
- Measures the expected variability of the sample mean.

Step 3

- T-Statistic
- Difference ÷ Standard Error
- A larger absolute t-value indicates stronger evidence against the null hypothesis.

T-Test Practice Question

A population has an **average IQ of 100**.

A team of researchers wants to investigate whether a new medication affects intelligence. They believe the medication may **increase, decrease, or have no effect** on IQ.

To test this, they randomly select **30 participants** who have taken the medication. The sample has the following characteristics:

- **Sample size (n): 30**
- **Sample mean IQ ($\bar{x}$) : 140**
- **Sample standard deviation (s) : 20**

Using a **95% confidence level ($\alpha = 0.05$)**, answer the following questions:

1. State the **Null Hypothesis (H_0)** and **Alternative Hypothesis (H_1)**.